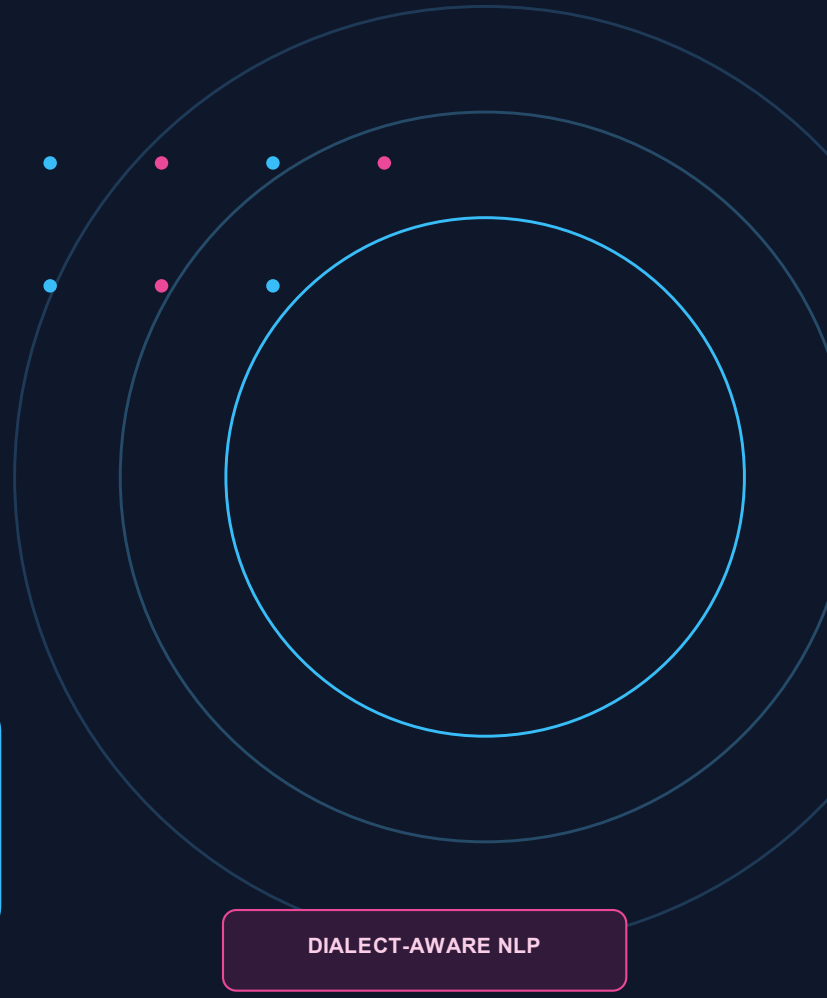
NLP GRADUATION PROJECT / 2026

End-to-End Fine-Grained Arabic Dialect Classification

Benchmarking Classical ML Ensembles vs. Pre-Trained Transformers AraBERTv02 & MARBERTv2



CHAMPION MODEL	UNSEEN TEST ACCURACY	MACRO F1
MARBERTv2	92.25%	92.24%

TARGET DIALECTS Beirut • Cairo • Doha • MSA • Tunis

GitHub / meedooatef22/arabic-dialect-classification

Why Is Arabic Dialect Classification Challenging?

Sociolinguistic Diglossia 01

- Formal MSA coexists with regional spoken dialects (Ammiya) used in everyday social media.
- No standardized spelling: Hamza (أ/إ/ا), Ya (ى/ي), and Ta-Marbuta (ة/ه) vary.

ORTHOGRAPHIC VARIATION

Heavy Morphological Agglutination 02

- Prepositions, negations, and pronouns fuse directly into word stems.
- Examples: “شورايك” (Beirut), “هالفيش” (Doha), “فماش” (Tunis) drive high OOV rates.

SUBWORD COMPLEXITY

Lexical Overlap & Short Text 03

- Neighboring dialects such as Beirut and Doha share vocabulary in unvocalized writing.
- Conversational messages are often only 2–4 words, providing minimal context.

LOW-CONTEXT AMBIGUITY

Dataset Overview & Strict Data Hygiene (MADAR Corpus)

Corpus & Split Integrity

RAW CORPUS	DEDUPLICATED	DUPLICATES
40,000	38,963	1,037

5 balanced classes • 8,000 raw sentences per dialect

Duplicates removed before splitting

Exact and post-cleaning deduplication prevents the same or normalized-equivalent sentence from crossing train/test boundaries.

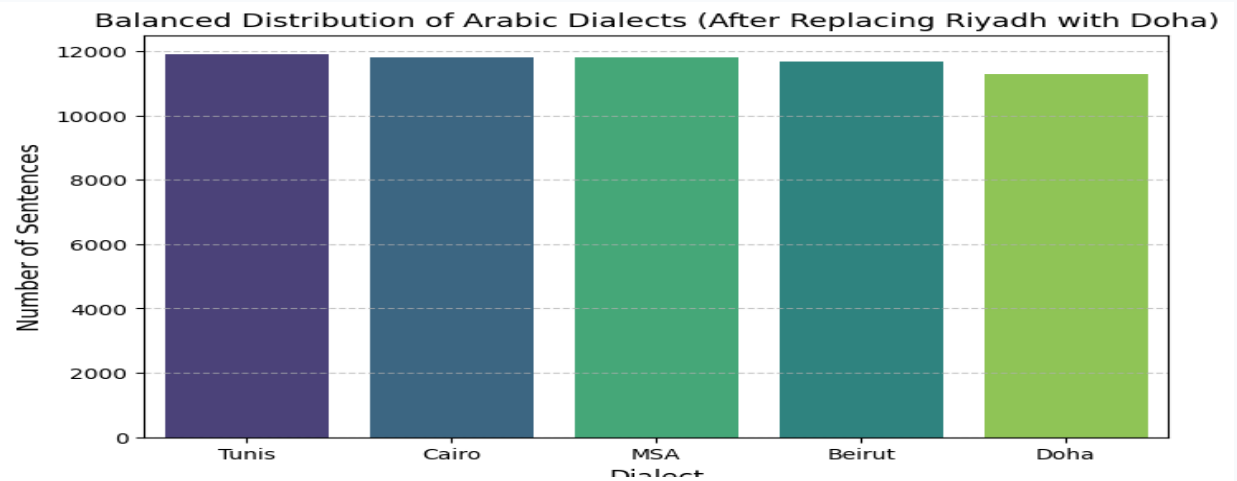
STRATIFIED 70 / 15 / 15 SPLIT

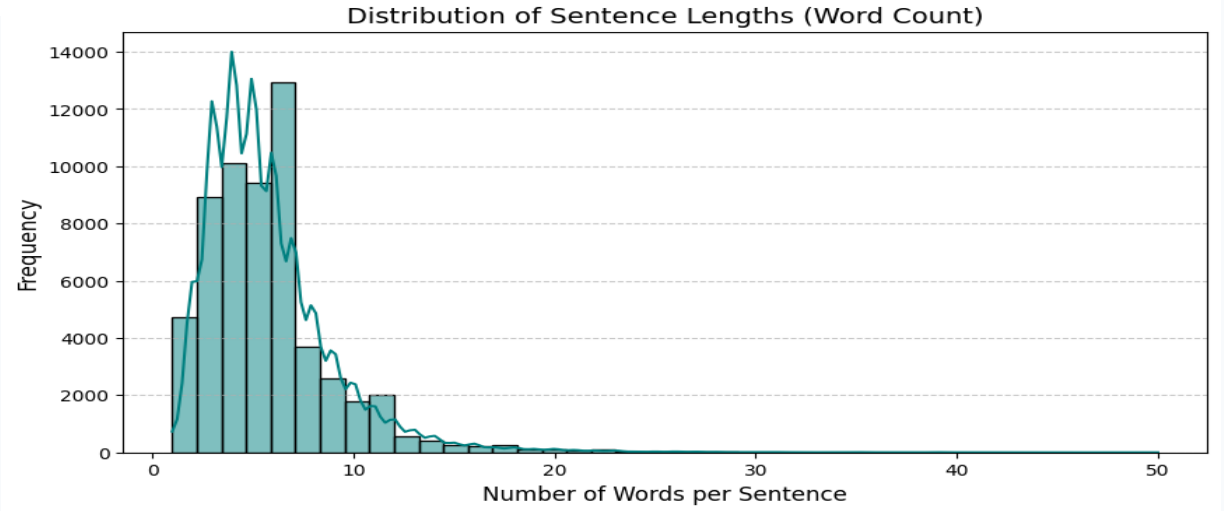


Train 27,274 (70%) **Validation 5,844 (15%)**

Unseen test: 5,845 sentences (15%)

Split after de-duplication • no intentional overlap





Domain-Specific Arabic Preprocessing Pipeline

01 Retain Arabic Stopwords

- Generic lists can delete negations, pronouns, and interrogatives.
- Here, function words are strong dialect fingerprints.

Cairo: مش/ ايه/ عايز• **Beirut:** شو/ ما/ بدي

Tunis: شنوة/ برشا/ فماش• **Doha:** وش/ مو/ ابغى

02 Retain Short Sentences

- One- and two-word turns may carry concentrated regional signals.
- Examples: “يعيشك”, “شورايك”, and “معلش”.

DON’T DISCARD SIGNAL

03 Deterministic Regex Normalization

clean_arabic_text

- Strip tashkeel, tatweel, URLs, digits, and non-Arabic characters.
- Normalize variants: (إ/ أ/ آ→ ا), (ى→ ي), (ة→ ه).

04 Before → After

RAW INPUT

“ شُورَأْيَك بـ 10دولار للتلاتة؟!!”

NORMALIZED OUTPUT

“شورايك ب دولار للتلاته”

End-to-End Modeling Progression

PHASE 01

Classical ML & Stacking

HYBRID TF-IDF • 150,000 FEATURES

- Word n-grams (1–3): 75k features for words and idioms.
- Character n-grams (2–5), char_wb: 75k features for subword patterns.
- LinearSVC, Logistic Regression, MultinomialNB, and SGD.
- 5-fold StackingClassifier meta-learner.

STRONG CLASSICAL BASELINE

PHASE 02

AraBERTv02 Fine-Tuning

aubmindlab/bert-base-arabertv02

PARAMETERS	EPOCHS
136M	4

- Pre-trained primarily on formal MSA news and encyclopedic text.
- Learning rate: 3e-5 • Batch size: 32.

MSA-ORIENTED PRE-TRAINING

PHASE 03 / CHAMPION

MARBERTv2 Fine-Tuning

UBC-NLP/MARBERTv2

PARAMETERS	DIALECTAL TWEETS
163M	1B

- Pre-trained on 1 billion dialectal Arabic tweets plus MSA data.
- 100k dialect-aware WordPiece vocabulary.
- Selected as the champion model for unseen-test performance.

DIALECT-AWARE PRE-TRAINING

Anti-Overfitting & Regularization Strategy in MARBERTv2

The Challenge / AraBERTv02

VALIDATION LOSS ROSE AFTER EPOCH 2



Training loss reached 0.072 by Epoch 4, while validation loss kept rising. Higher learning rates encouraged memorization of short phrases.

The Solution / MARBERTv2

2e-5	Peak learning rate
10%	Linear warmup
0.05	Weight decay
0.05	Label smoothing

TRAINING CONTROLS

- EarlyStoppingCallback: patience = 2; monitor eval_f1_macro.
- load_best_model_at_end = True • FP16 mixed precision.

Experimental Benchmark Results Across All Models

CLASSICAL STACKING / TEST ACCURACY

90.93%

4-model meta-learner

ARABERTV02 / VALIDATION ACCURACY

91.03%

Fine-tuned transformer

MARBERTV2 / UNSEEN TEST ACCURACY

92.25%

Macro F1: 92.24%

Model Architecture	Feature Type	Val Accuracy	Test Accuracy	Test Macro F1
Logistic Regression	Hybrid TF-IDF (150k)	89.37%	89.70%	89.70%
SGD Classifier	Hybrid TF-IDF (150k)	89.90%	90.20%	90.20%
Multinomial Naive Bayes	Hybrid TF-IDF (150k)	90.35%	90.47%	90.45%
LinearSVC (C=0.5)	Hybrid TF-IDF (150k)	90.59%	90.52%	90.52%
Classical Stacking Ensemble	4-Model Meta-Learner	90.85%	90.93%	90.92%
AraBERTv02 (Fine-Tuned)	Subword Transformer	91.03%	—	91.00%
MARBERTv2 (Champion)	Dialect-Aware Transformer	92.02%	92.25%	92.24%

All figures shown as reported in the project benchmark; "—" indicates no test accuracy listed for AraBERTv02.

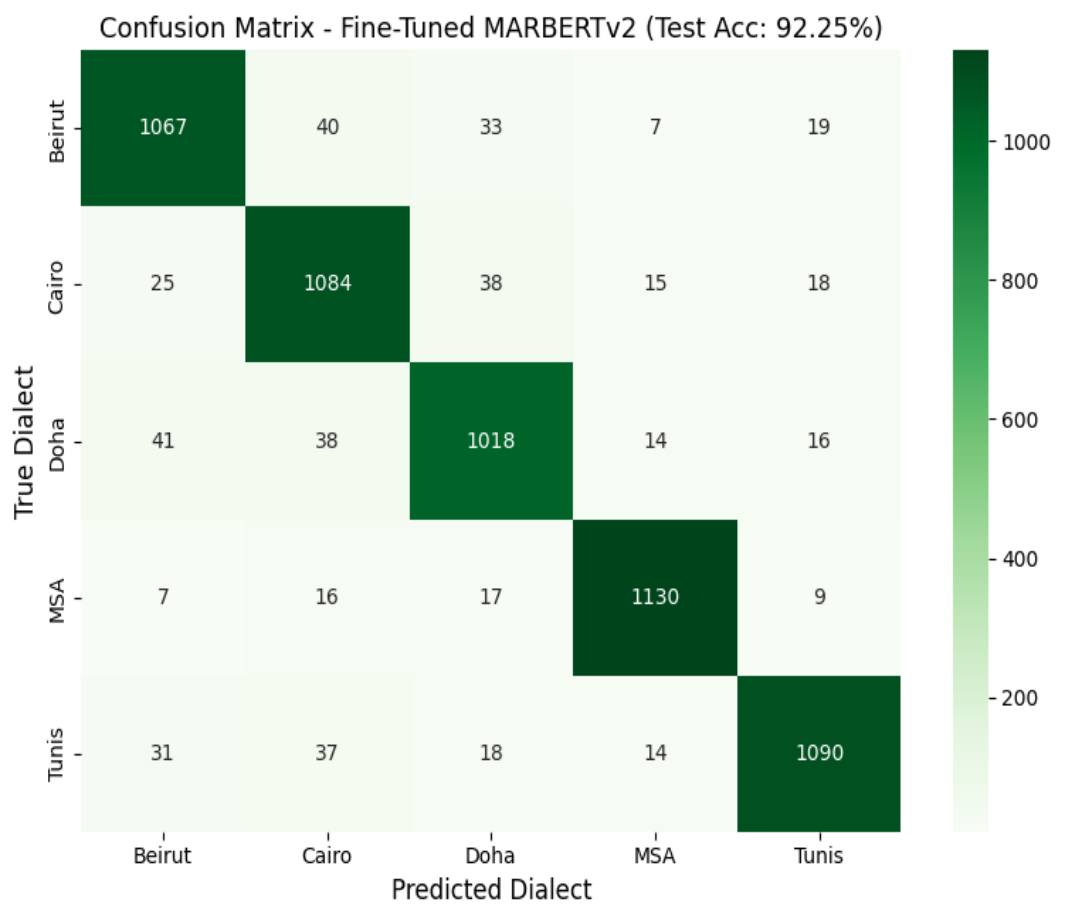
Champion Model (MARBERTv2) Per-Dialect Breakdown

Dialect	Precision	Recall	F1-Score	Support
Tunis (Maghrebi)	95.21%	93.69%	94.44%	1,173
MSA (Formal)	93.60%	94.61%	94.10%	1,168
Cairo (Egyptian)	92.42%	91.47%	91.94%	1,172
Doha (Gulf)	90.02%	91.47%	90.74%	1,172
Beirut (Levantine)	90.09%	89.99%	90.04%	1,160

77 ERRORS ELIMINATED

530 → 453

14.5% fewer test errors on the same test set



Deep Error Analysis: Why Do 7.75% (453) Errors Remain?

Average Sentence Length

7.21 words
Correct predictions
5,392 samples

3.86 words
Misclassified
453 samples

Average Prediction Confidence

95.83%
Correct predictions

77.57%
Misclassified • −18.26 pp

Theoretical Dataset Ceiling / Ambiguous Written Language

TOP CONFUSION PAIR

Beirut ↔ Doha

91 mutual errors; shared Levantine/Gulf vocabulary in unvocalized text.

EXAMPLE 1 / A MODEL-RESCUED CASE

“شورايك بعشره دولار للتلاته”

Stacking → Doha | MARBERTv2 → Beirut (97.92% confidence)

EXAMPLE 2 / NATURAL AMBIGUITY

“ايوه مسموح” • “تاني تقاطع عاليمين”

Everyday phrases can occur in Cairo, Beirut, and Doha; text alone may not resolve the label.

Production Web Platform (Streamlit & Ngrok Deployment)

Three Modes in the Web App

01

Single Sentence Analysis

Normalize text, return top dialect + confidence, and show probability bars for all five classes. Smart linguistic alert when confidence <45%.

02

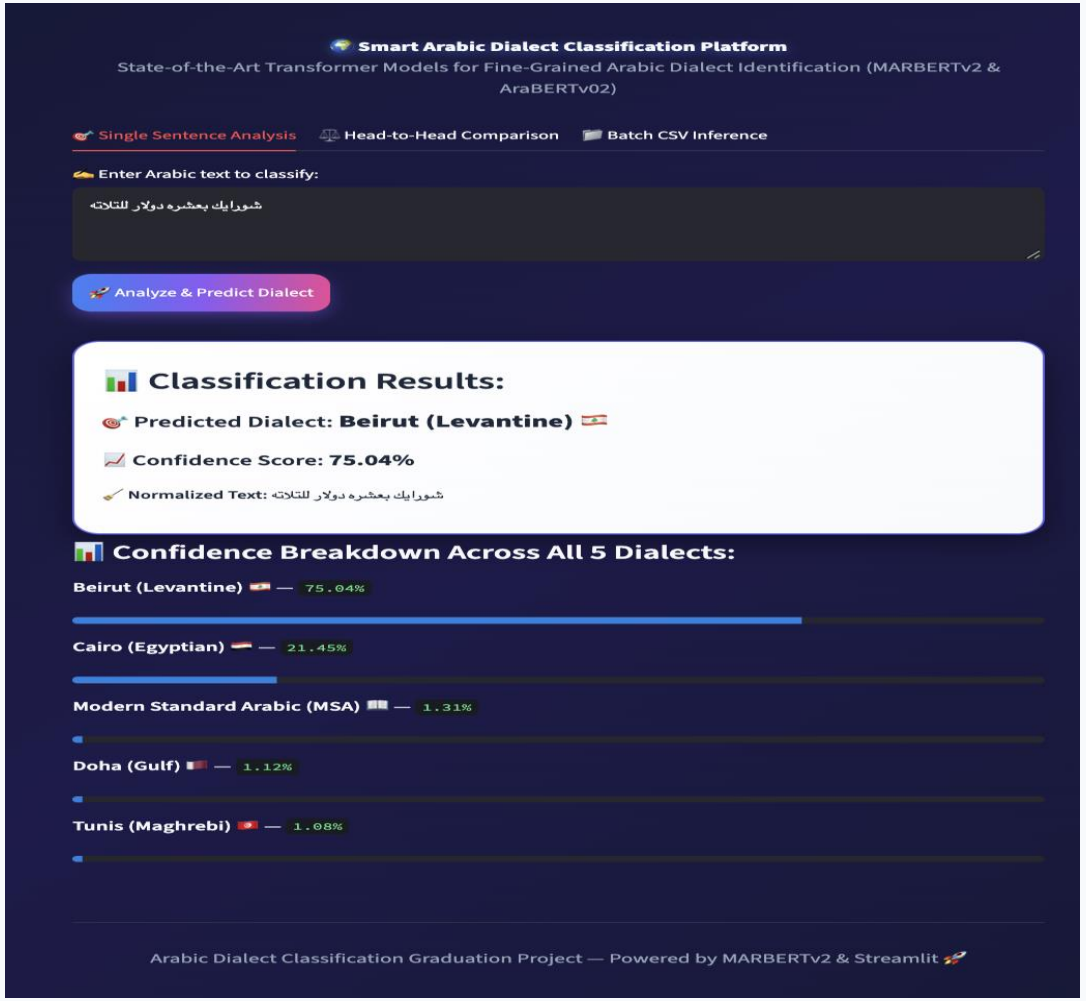
Head-to-Head Comparison

Run MARBERTv2 and AraBERTv02 side by side for a live comparison (92.25% vs. 91.03% reported benchmark).

03

Batch CSV Inference

Upload a CSV with a `text` column, classify hundreds of rows on GPU, then download UTF-8 results.



Conclusion & Future Directions

What We Achieved

01 — Leak-free data pipeline

38,963 unique sentences after deduplication and stratified splitting.

02 — Best-performing model

MARBERTv2 reached 92.25% test accuracy, ahead of AraBERTv02 (91.03% validation) and Stacking (90.93% test).

03 — Open-source delivery

GitHub documentation, README.md, six modular notebooks, technical report, and bilingual Streamlit apps.

Future Work

01 — Broader dialect coverage

Expand from five target dialects to 25+ fine-grained Arab cities (MADAR-26).

02 — Audio + text fusion

Use pronunciation to disambiguate very short phrases with identical spellings.

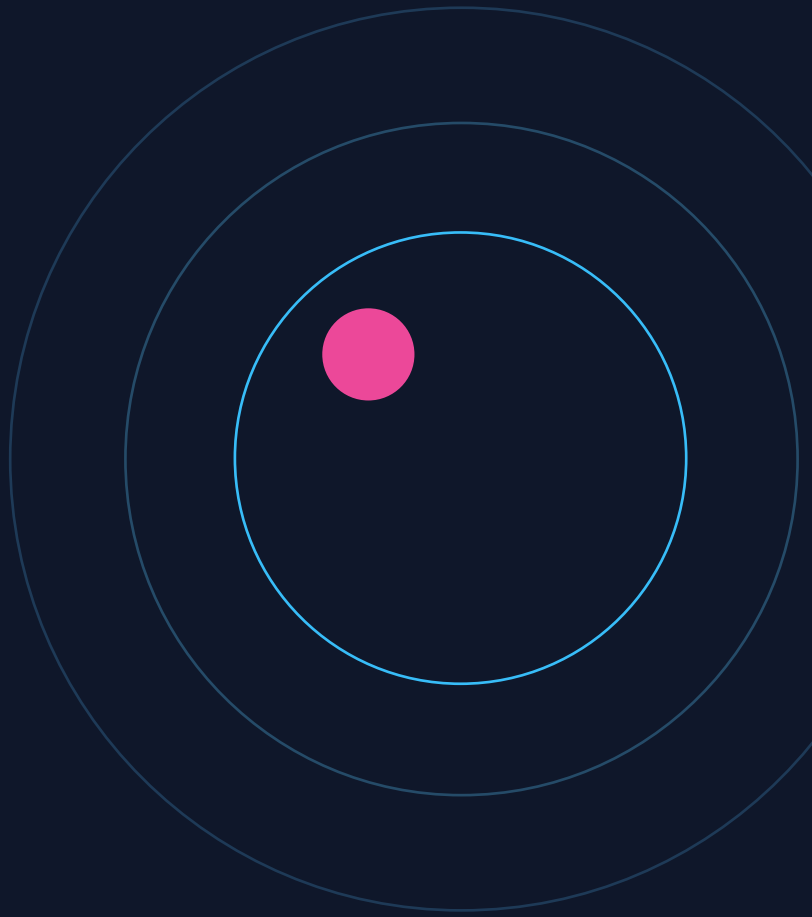
03 — Efficient edge inference

Quantize with ONNX / INT8 for low-latency mobile and edge deployment.

FROM DATA HYGIENE → MODEL BENCHMARKING → PRACTICAL DEPLOYMENT

Thank You!

Live Interactive Demo & Q&A Session



READY FOR LIVE TESTING ON STREAMLIT

Single Sentence | MARBERTv2 vs. AraBERTv02 | Batch CSV

GITHUB REPOSITORY

github.com/meedooatef22/arabic-dialect-classification